

Chloe Kim

Resume – Academic Emphasis

📍 Cranbrook Kingswood Upper School, 39221 Woodward Ave. Bloomfield Hills, MI

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Academic Interests & Selected Outputs

Focus: AI × Human Decision-making under Uncertainty & Fairness

- Cognition & intelligence — perception, reasoning, risk
- Applied domains: social applications of AI, AI insurance, legal networks, healthcare triage

Keywords: Computational cognitive science; reinforcement learning & Bayesian inference; NLP; causal inference

Selected Outputs

- *Professional Academic Presentations* (3): EURO 2025; Helmholtz AI 2025 (*Spotlight*)
— Human *vs.* model-based reasoning; Recognition & uncertainty, fairness; AI applications
- *Research Papers* (2): PLOS One; Cognitive Systems Research (submitted)
— Emergent architecture of law; Intuition *vs.* semantic reasoning by humans and AI
- *Patents* (2): AI insurance framework; Influence scoring of judicial decisions (Sole Inventor)
- *Software Reg.* (4 works): Network dynamics, power-law & inequality study (Sole Developer)

Education

Cranbrook Kingswood Upper School *High school student* Aug 2022–Jun 2026 (expected)
Bloomfield Hills, MI

Harvard University *Undergraduate-level studies* July 2024–Aug 2024
Cambridge, MA

Molecular Biology (BIOS-S1a) (A-) and Historical Linguistics (LING-120) (A)

Studies at Harvard Summer School (for credit)

南京时代双语学校 (Nanjing Times College) *Middle school student* Apr 2020–Jun 2022
南京(Nanjing), China

경희초등학교 (Kyunghee School) *Elementary school student* Mar 2014–Feb 2020
Seoul, Republic of Korea

Selected Research & Intellectual Property

Patents *Sole Inventor*

“System and method for evaluating AI output consistency and providing dynamic insurance”
#10-2025-0104581, patent pending (Korea)

- Quantifies AI risks (factuality, logical/semantic consistency, bias) into actuarial metrics.
- Designs dynamic premium pricing & claim validation linked to AI performance.
- *AI Insurance* – recompensation for AI hallucinations

“Systems and Methods for Temporal Influence Scoring of Judicial Decisions”

#10-2025-0166979, patent pending (Korea)

- Builds a dated, directed legal-citation index; computes influence measures (adoption/impact).
- Improves retrieval via structure-weighted rankings and rolling-window alerts that surface pivotal precedents and enable monitoring.

Software Registrations *Sole Developer*

“Interpreting Social Intent and Behavior: Temporal Dynamics of Network Heatmaps” (#035490)

- Analyzes temporal networks, creating centrality- & time-based maps of social relations.

“Analysis of Social Intentionality: Visualization of Case-law Networks” (#035307)

- Processes legal datasets, computing centrality & visualizing case-law structures.

“Simulation of Behavioral Relationships in Power-law Dynamics using AI” (#035489)

- Models behavior relations under power-law dynamics with AI sampling & visualization.

“Inequalities in Human Behavior: A Graph-based and AI Interpretation” (#034662)

- Analyzes behavioral inequality using AI/graphs, quantifying disparities via references.

Research Presented & Papers *First author & Presenter*

“Dynamic Network Analysis of Legal Citation Networks (especially medical)”

34th European Conference on Operational Research (EURO 2025), Leeds, UK

- Analyzed 86,000+ Supreme Court cases using network and power-law analysis, showing concentrated reliance on landmark precedents.
- Illustrates how structural citation patterns shape legal & medical decision-making.

“Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)”

PLOS One: under review (Oct 2025)

“An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI”

Helmholtz AI Conference 2025, Karlsruhe, Germany

- Compared 40+ multilingual participants with GPT-4 on sound-to-shape tasks, revealing humans’ intuitive mappings *vs.* AI’s semantic reasoning.
- Findings support dual-mode cognition, highlighting differences in human and machine decision-making.

“Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination”

Cognitive Systems Research: under review (Dec 2025)

“Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction” (*Spotlight*)

Helmholtz AI Conference 2025, Karlsruhe, Germany

- Trained on 500+ patient cases with fused imaging (Vision Transformer) and clinical features (XGBoost), achieving 92.7% accuracy and AUROC 0.90.
- Results demonstrate scalable, interpretable AI for airway management, highlighting contrasts between human judgment and standardized decision-making.

Work Experience

Lawfirm Hoam *Research Assistant*

*Summer 2024–
Seoul, Rep. of Korea & Remote*

- Conduct data (legal) analytics and case research using AI and quant. network analysis.
- Presented results at an academic conference.

Quilt.ai *Research Intern*

*Jul 2024–Dec 2024
Boston, MA & Remote*

- Evaluated and researched AI narrative generation for fashion ads that support human perception and cognition.
- Ran market surveys; produced insight memos used by the design/product teams.

Media art Yeseung Lee's team *Assistant Editor/Supporter*

*Dec 2024–
Seoul, Rep. of Korea & Remote*

- Supported a project on art and human cognition.
- Edited artist statements and exhibition texts (Singapore, Seoul Dongdaemun Design Plaza (DDP), National Museum of Modern Art (MMCA) exhibitions).

Elroi AI *Research Intern*

*May 2025–
Troy, MI & Remote*

- Prototyped and studied modules for generation & HAI workflows; conducted user tests.
- Delivered a technical brief.

Herring Insight *Volunteer Analyst*

*July 2024–
Newton, MA*

- Analyzed cultural-difference on how to support deprived people's education.

SAT Mathematics Tutoring *Tutor*

*June 2025–September 2025
NYC, NY & Remote*

- Tutored high school students in SAT Math preparation and strategies.

Computer Science Projects *Independent researcher*

*Jun 2022–
Bloomfield Hills, MI & remote*

- Built coding projects including BMI Calculator, Quiz Game, Server-Client Trading Apps, and AI&ML-based game simulations. (9th – 12th grade)

Leadership & Community Service

Varsity Rowing Team *Coxswain*

- Coach/manage boats (steering/making calls & corrections), mentor novice rowers, delegate roles, manage race day logistics & encourage team support

Linguistics Club *Founder, Co-President*

- Founded Linguistics Club to promote different branches of linguistics; raise awareness of diverse languages; NACLO prep; AI+ Linguistics research

Crane Clarion (School newspaper) *Assoc. Arts Editor, Graphic Designer, Staff Writer*

- Led artists & writers; mentored junior writers; established journalistic protocols; publicized arts & school spirit; wrote cultural sens. guidelines

Ethics Bowl Club *President/Founder*

- Founded club; recruited, hosted state-wide practice events, organized spaces for ethical thinking & dialogue in & out of school & community

LITE (*Leaders in the Environment*) *Food Board Captain*

- Liaison to school meal providers; promoted inclusive meals; systemized negotiation process btw. student body, faculty, and providers; org 10+ events

Cranbrook Kingswood Chamber Orchestra, Thornlea Quartet *Violinist; Organizer*

- Violinist in school orchestra; performed at school/community; organized fundraising concerts for Brilliant Detroit; mentored peers

Tutoring *Comm. Service – Volunteer tutor*

- Independent & Volunteer – Local & Overseas; hybrid—in-person/online via Zoom/Teams)
- Tutored 5-7 under-resourced students: English (Chinese & Korean) + SAT Math (US); ran Zoom lessons; built practice sets; measurable score gains

Gold Key *Sophomore Lead***“Ergasterion” – Theater & Drama club** *Stage Crew & Member***Student Council** *Member***Awards and Honors**

National Latin Exam Cum Laude and 2x Magna Cum Laude – G9, 10, 11; *National*

Ethics Bowl Central Division 4th place – G10, 11; *State*

3rd in State for National Economics Challenge – G10; *State*

National Merit semifinalist – *National*

District MSBOA 1st Division (solo) & 1st/2nd Division (quartet) – G9, 10, 11; *State*

NEC – National Economics Challenge; 3rd (D. Ricardo) – G10; *State*

Strickland Writing Award – G9; *School*

Dean’s List & Honors – G9, 10, 11; *School*

AP Scholar with Distinction Award – G10; G11; *National*

Skills

AI and Data Analytics

- (1) Statistics and Data Analytics + (2) Network and Complex System Analysis

Technical Computer Science: *Java, Python, Julia* – Applications to Semantics and Philosophy

Additional

Languages

English (*native*); Korean (*native/bilingual*); Chinese (Mandarin, *native*); French (*intermediate*); Japanese (*intermediate*); Latin (*NLE Latin III*)

Training (selected external coursework)

MIT OCW (Python/Julia, abstraction, object-oriented programming, recursion); FastCampus ML (TensorFlow, Docker, LLM agents); Human Asia (“*Human Rights*” for North Korea)

Citizenship: United States

Abstracts

Patents

- System and method for evaluating AI output consistency and providing dynamic insurance (#10-2025-0104581, *Pending*)

Abstract translated from the filed application

- *Abstract*

This invention relates to novel systems and methods that address the reliability issues of outputs generated by artificial intelligence such as large language models (LLMs) and that provide insurance coverage for the associated risks.

The invention organically integrates an “AI Output Consistency Evaluation Module,” which assesses the reliability of AI outputs across multiple dimensions, with an “AI Output Insurance System,” which operates insurance products based on the evaluated risks. The evaluation module comprehensively verifies factors such as factual accuracy, logical and semantic consistency, and bias to generate a quantitative composite risk profile.

The insurance system directly ingests this profile and dynamically calculates and adjusts premiums and coverage terms in line with the model’s performance; in the event of an incident, it verifies the validity of claims based on pre-recorded, objective performance data.

Through this approach, the invention converts the unpredictable technological risks of AI into manageable insurance liabilities and promotes the responsible adoption and use of AI by enterprises.

- Systems and Methods for Temporal Influence Scoring of Judicial Decisions
(#10-2025-0166979, *Pending*)

- *Abstract*

A computer-implemented system generates time-scoped influence signals for legal search and monitoring by dynamically analyzing citation networks of judicial decisions. It ingests a large corpus with citation metadata, normalizes references, and builds a dated, directed citation index. On this index, the system computes two complementary measures for each decision: an adoption index (in-degree-based reliance by later cases) and an impact index (recursively computed from citation relations). Scores are evaluated in rolling time windows and can be linked to subject labels to produce interpretable timelines and dashboards. Outputs include structure-weighted rankings, early surfacing of pivotal precedents, and alerts when influence moves across topics or time bands, with rationales that show contributing sources and time weights.

Research Presented (in Conferences)

- Dynamic Network Analysis of Legal Citation Networks (esp. medical) in South Korea (1951–2023)

34th European Conference on Operational Research (EURO)

- Abstract

We analyze approximately 86,000 legal cases from South Korea (1951–2023) by constructing (complex) citation networks, modeling each case as a node and citation relationships as edges. Through centrality/hub analysis, we identify key hub cases. We extract and analyze a focused subnetwork comprising medical-related cases.

Segmenting the network along presidential cycles reveals that the distribution of medical hub cases changes over time. Cases related to criminal liability in medical accidents were prominent hubs from the late 1990s to early 2000s, whereas subsequent administrations saw civil rulings on patients' rights and institutional responsibilities become new hubs. This analysis illustrates structural reorganization within the network.

From an optimization perspective, we apply the minimum dominating set (MDS) approach, supplemented by AI-assisted heuristics, to extract a minimal influential subset within the medical subnetwork. Complexity analysis shows power-law degree distributions with the exponent approximately 2.1, indicating scale-free characteristics. An AI-assisted optimization, balancing sparsification and policy relevance, facilitates a more targeted identification of legally and policy-relevant cases.

We suggest quantitative investigations into the structural features and dynamic evolution of legal citation networks in the medical domain. These provide practical implications for legal and policy decision-making in healthcare governance.

- How I obtained the opportunity

I secured the opportunity through direct outreach and an interview with Attorney Seongwoong Kim (Lawfirm Hoam); I collaborated under his mentorship. My research was accepted and presented at the conference.

- Primary author (if not me)

I am the first author and presenter.

- Scholarly Guidance and Oversight

Seongwoong Kim – Mentor, Co-author

- An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI

Helmholtz AI Conference 2025

- *Abstract*

We experimentally analyze how association stimuli are processed through distinct cognitive mechanisms by humans and AI (large language models, LLMs) in visual generation tasks based on linguistic association. As stimuli, we employ nonsymbolic sound words (non-symbolic sounds), which represent meaningless but structured sound sequences. This study focuses on the contrast between human intuition and LLMs' learned semantic reasoning. Humans tend to visualize 'sound-associated words' without prior learning, relying on immediate sensory and affective cues to produce intuitive forms—curves or pointed shapes. In contrast, LLMs such as GPT map these words onto shapes based on large-scale text-driven knowledge, using semantic structures and statistical patterns.

To verify this, we conduct experiments with both a diverse cross-linguistic and cross-cultural human participant group ($n \approx 40$; including Korean, Chinese, English, French, and Hindi speakers) and an LLM (GPT-4). Shape categorization classifies the imagined (in the case of humans) and generated images (in the case of the LLM) into basic geometric forms such as circles, triangles, and rectangles. Results show that humans rely on pre-existing intuitive mappings between sound and shape, performing direct sensory-to-visual transfers without explicit learning. By contrast, LLM outputs (expression of shapes) are generated through prior semantic knowledge learned from language corpora, reflecting a meaning-driven reasoning process.

We identify a fundamental gap between human intuition-based learning and LLMs' meaning-based learning. These findings enhance our understanding of multimodal perception and generative mechanisms in foundation models (fundamental AI models).

- *How I obtained the opportunity*

This is an independent research project I conceived and executed solo. My research was accepted and presented at the conference.

- *Primary author (if not me)*

N/A — I am the only author.

- *Scholarly Guidance and Oversight*

None (independent/self-directed)

- Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction (*Spotlight*)

Helmholtz AI Conference 2025

- *Abstract*

The Mallampati score is widely used to assess the difficulty of intubation and endoscopic procedures but suffers from high inter-rater variability. We present an AI model that automatically classifies Mallampati scores and predicts the need for endoscopic assistance using multimodal data. A dataset of 500 patients from a hospital in Korea was used, which includes Mallampati photographs and clinical variables. A Vision Transformer (ViT) processes imaging data, and gradient boosting is applied to clinical features. These AI models are fused to predict Mallampati classes (I–IV) and the binary outcome of endoscopy necessity. Additionally, we integrate a large language model (LLM) to generate clinically interpretable explanations to support decision-making. Our model achieves an accuracy of approximately 92.7% in Mallampati classification and an AUROC of 0.9 for endoscopy necessity prediction. This highlights the potential of deep learning interpretability to standardize airway assessments and support procedural planning.

- *How I obtained the opportunity*

Secured via direct outreach to Research Professor Haneol Cho (Sejong University); joined as a research collaborator with his support and access to his lab's dataset. My research was accepted for a spotlight presentation which was carried out at the conference.

- *Primary author (if not me)*

I am the first author and presenter.

- *Scholarly Guidance and Oversight*

Dr. Haneol Cho – Collaborator, Supporter

Research Papers

Full manuscripts are provided as a separate PDF.

Abstracts are included below for your reference.

- Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)

PLOS One (an academic journal), Manuscript Submitted for Review (with S. Kim)

- Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination

Cognitive Systems Research (an academic journal), Manuscript Submitted for Review

- Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)

PLOS One (an academic journal) (with S. Kim)

- *Abstract*

The evolution of a nation's legal system can be read as a data-driven and AI narrative of its socio-political development. Conceptualizing law as a complex adaptive system, we conduct the first large-scale dynamic network analysis of the Republic of Korea's Supreme Court jurisprudence, mapping 86,000+ decisions from 1951–2023 into a comprehensive citation network. The network is not random but scale-free, with a power-law distribution ($\alpha = 3.29$) consistent with preferential attachment, and exhibits substantial inequality (Gini coefficient of 0.604), whereby a small number of landmark cases operate as influential hubs. Crucially, we distinguish two modes of precedent influence: popularity (in-degree) versus authority (centrality). Highly cited cases are not always the most authoritative; popular precedents often serve as routine tools, whereas high authority cases constitute the backbone of legal reasoning. A focused examination of the medical-law sub-network reveals an even more hierarchical structure ($\alpha = 2.14$), indicating that knowledge specialization amplifies structural inequality. A temporal segmentation by presidential administrations uncovers a marked semantic shift in influential medical jurisprudence: from state-centric criminal enforcement in earlier periods to citizen-centric civil and administrative law emphasizing patient rights and professional accountability in later democratic eras. Viewed through a cognitive lens, the citation network functions as a collective mind: judicial behavior, which is driven by bounded cognition and heuristic search, produces a brain-like architecture that stores institutional memory and channels inference through authoritative hubs. This dual structural-semantic account offers a behavioral and cognitive framework for understanding how legal doctrine co-evolves with societal change, with implications for legal theory, judicial behavior, policy design, and the development of artificial intelligence in law.

- Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination

Cognitive Systems Research (an academic journal)

- *Abstract*

Humans can intuitively associate phonological forms with visual properties, exemplified by sound symbolism effects such as “*bouba-kiki*,” where people consistently match certain pseudowords with particular shapes. In contrast, AI large language models (LLMs) like GPT-4 learn from text and may rely on learned semantic and distributional regularities rather than perceptual intuition. We compare humans and an LLM in a shape imagination task using non-symbolic stimuli: six novel nonwords (Nonword A–F). Forty multilingual participants ($n = 40$) read each nonword, imagined its pronunciation, and depicted a simple shape, while GPT-4 received the same strings as text input and generated a single shape description per stimulus. Human responses showed consensus consistent with intuitive sound-symbolic mapping. GPT-4, by contrast, produced shapes that diverged from the human majority, suggesting a reliance on language-based associations rather than pre-linguistic crossmodal intuition. The resulting mismatch highlights a cognitive gap between human intuitive processing and the LLM’s semantic reasoning, and provides empirical evidence relevant to the symbol grounding problem. Our findings motivate future work on perceptual grounding and multimodal learning in foundation models.

Papers

Research Papers

- Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)

PLOS One (an academic journal), Manuscript Submitted for Review (with S. Kim)

- Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination

Cognitive Systems Research (an academic journal), Manuscript Submitted for Review

Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)

Chloe Kim^{1,*}, Seongwoong Kim^{2,*}

¹ Cranbrook Kingswood Upper School, Bloomfield Hills, MI, USA

² Lawfirm Hoam, Seoul, South Korea

* Correspondence should be addressed to ckim26@cranbrook.edu (CK) and iamsmile100@gmail.com (SK).

Abstract

The evolution of a nation's legal system can be read as a data-driven and AI narrative of its socio-political development. Conceptualizing law as a complex adaptive system, we conduct the first large-scale dynamic network analysis of the Republic of Korea's Supreme Court jurisprudence, mapping 86,000+ decisions from 1951–2023 into a comprehensive citation network. The network is not random but scale-free, with a power-law distribution ($\alpha = 3.29$) consistent with preferential attachment, and exhibits substantial inequality (Gini coefficient of 0.604), whereby a small number of landmark cases operate as influential hubs. Crucially, we distinguish two modes of precedent influence: popularity (in-degree) versus authority (centrality). Highly cited cases are not always the most authoritative; popular precedents often serve as routine tools, whereas high authority cases constitute the backbone of legal reasoning. A focused examination of the medical-law sub-network reveals an even more hierarchical structure ($\alpha = 2.14$), indicating that knowledge specialization amplifies structural inequality. A temporal segmentation by presidential administrations uncovers a marked semantic shift in influential medical jurisprudence: from state-centric criminal enforcement in earlier periods to citizen-centric civil and administrative law emphasizing patient rights and professional accountability in later democratic eras. Viewed through a cognitive lens, the citation network functions as a collective mind: judicial behavior, which is driven by bounded cognition and heuristic search, produces a brain-like architecture that stores institutional memory and channels inference through authoritative hubs. This dual structural-semantic account offers a behavioral and cognitive framework for understanding how legal doctrine co-evolves with societal change, with implications for legal theory, judicial behavior, policy design, and the development of artificial intelligence in law.

1 Introduction

Modern judicial systems are often viewed as static hierarchies of rules, but in reality they form dynamic, interconnected, and evolving ecosystems of precedents.

Complexity theory offers a powerful framework for understanding law as an emergent, self-organizing system in which an interactive network of actors and institutions

produces order without any central guiding hand [1–3]. In such a complex adaptive system, legal rulings do not exist in isolation; rather, each judicial opinion builds upon or distinguishes prior decisions through a web of citations. These citation links weave a living network that serves as the memory and structural backbone of the law. Over more than 70 years of South Korean legal history, this network of precedents has continually evolved, reflecting how thousands of individual judicial decisions collectively shape the “*mental and cognition model*” of the law.

Legal citation network analysis provides a quantitative lens to empirically map and measure this complex system [4,5]. In a citation network, individual cases are represented as nodes, and a directed edge from Case A to Case B indicates that A cites B. By examining the topology of this network, one can assess the importance of a case not just by how often it is cited (its direct popularity), but by its position in the overall structure (its authority *within* the network). We operationalize these two modes explicitly: popularity as in-degree and authority as centrality. Each citation can be viewed as a latent judgment by a judge about the relevance of a prior case to a current dispute [6,7] The aggregation of these millions of citation judgments forms a self-organizing web of legal knowledge – a collective reasoning process that constitutes a crucial part of the law itself as well as the cognitive and behavioral decisions.

Although network analysis has been applied to jurisprudence in some contexts (notably the U.S. Supreme Court [8,9]), the South Korean court system remains largely uncharted from this perspective. Prior work has emphasized text-mining tasks rather than dynamics of influence. This study addresses that gap by constructing and analyzing a dynamic citation network encompassing 86,161 Supreme Court cases from 1951 to 2023. Our research makes three primary contributions: First, it characterizes the macro-scale architecture of South Korea’s legal precedent network, rigorously testing for non-random (scale-free) properties. Second, we perform an in-depth analysis of a particular domain —medical law— where precedents are expected to significantly influence societal norms around healthcare and professional responsibility. Third, it tracks the structural and semantic evolution of this sub-network across distinct political eras, providing a unique window into the interplay among legal doctrine, societal change, and human judicial cognition over time.

By treating the web of legal citations as observable data, we reframe jurisprudence as not only a philosophical or interpretive discipline but also an empirical and computational one. The citation network represents the macroscopic output of collective cognitive and linguistic processes. It is essentially the *emergent mind* of the

law, arising from the interactions of innumerable judicial decisions. Analyzing this network’s architecture is akin to conducting a form of cognitive science at the societal level, allowing us to investigate how complex group behavior (here, the judicial decision-making process) gives rise to the evolving intelligence of the legal system itself, including psychological attention and societal dynamics.

We describe the data and network construction, explain the analytical measures, and present the structural and temporal–semantic results. We then examine the medical sub-network to highlight specialization. Implications for cognition and mind, policy, and AI are discussed; the paper concludes.

2 Approaches and Methods

2.1 Data and Network Construction

We assemble a corpus of 86,161 Supreme Court of Korea decisions spanning 1951–2023 from a law firm–affiliated commercial legal database of South Korean court judgments. Each record contains a unique case identifier, decision date, full-text opinion, case type (criminal, civil, administrative, and others), and the case IDs of precedents cited by the decision. We programmatically de-privatize and parse the corpus to standardize metadata and citation strings, and to construct the citation graph used in our analyses.

Using the cleaned citation data, we model the legal corpus as a directed graph $DG = (V, E)$, where V is the set of unique cases (nodes) and a directed edge $(u, v) \in E$ exists if case u cites case v in its written opinion. In this representation, edges point from the citing case to the cited case, thereby capturing the historical flow of influence from earlier to later jurisprudence. Cases with no connections (neither citing nor cited by any other case in the corpus) are excluded as isolated nodes.

Within this network we distinguish two complementary perspectives: the *referring* (outgoing citations from a case, out-degree) perspective, which counts how many precedents each case cites, and the *referred* (incoming citations to a case, in-degree) one, which counts how often each case is cited by later cases. Our primary analyses focus on the referred/in-degree network as a representation of precedent influence, since a case’s in-degree serves as a direct proxy for its impact on subsequent legal reasoning and future rulings.

We analyze a domain-specific sub-network of the network focusing on *medical law*. To construct this medical law sub-network, we filtered the dataset for cases explicitly related to medical issues. This is done by filtering the corpus for cases whose titles or

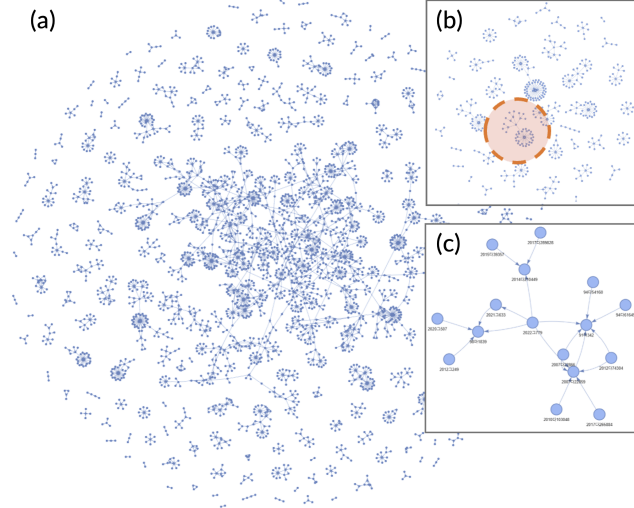


Fig 1. Comprehensive Legal Network (Supreme Court of Korea, 1951–2023). The directed citation graph $DG = (V, E)$ is constructed from cleaned case-to-case citations; a directed edge $u \rightarrow v$ is present when case u cites case v in its written opinion. Isolated cases (degree 0) are excluded, and disconnected components remain. (a) Global view of the network. (b) Zoomed region (dashed circle) highlighting hub-like precedents in the *referred* perspective. (c) Local neighborhood of a representative precedent showing incoming citations from later cases. Consistent with our study design, analyses focus on the referred perspective as a proxy for precedent influence.

opinion texts contain predefined Korean medical-related keywords (e.g., Medicine, Doctors, Medical Service Act, Medical Law, malpractice, negligence, and patient). The resulting sub-network is then treated as an independent network for structural analysis, allowing us to compare its properties to the comprehensive network.

2.2 Analytical Measures

To probe the structure of the citation network, we utilize measures from complex network theory. We compute the analytics to characterize the structure:

- **Degree Distribution and Power-Law Fitting:** For each node (case) we obtain the in-degree. After we fit in the complementary cumulative distribution function, then plot the probabilistic distribution function (PDF) of the in-degrees on a log-log scale. With a lower bound cutoff to focus on the tail of the distribution where power-law behavior is observed, we compute the exponent estimate α in the power-law distribution $P(x) \propto x^{-\alpha}$.
- **Gini Coefficient:** To quantify inequality in citation counts, we generated a Lorenz curve for the distribution of in-degrees and compute the Gini coefficient G . The Gini is calculated as the ratio of the area between the Lorenz curve and the

line of equality, over the total area under the line of equality. The higher coefficient G , the more unequal, in which all citations go to one case. Thus it provides a summary of how unevenly citations are spread among cases.

- **Authority:** We obtain the authority based on centrality, PageRank score $PR(v)$. PageRank, originally developed to rank web pages [8,10], is a recursive measure of influence that takes into account not just the number of citations a case has, but who is citing it. We compute $PR(v)$ for each case v using the standard iterative formula:

$$PR(v) = \frac{1-d}{N} + d \sum_{u \in M(v)} \frac{PR(u)}{L(u)}, \quad (1)$$

where $M(v)$ is the set of cases that cite v , $L(u)$ is the number of outbound citations from case u , N is the total number of cases, and d is the damping factor (0.85). Iteration continues until convergence. A case earns a higher PageRank score if it is cited by other cases that are themselves highly cited or authoritative. This allows us to distinguish between merely well-cited cases and those that sit at the core of the network’s structure. We also consider simple in-degree. By ranking cases on both metrics, we can contrast popular precedents (high in-degree) with authoritative precedents (high centrality), highlighting differences between cases that are widely cited and those that serve as essential hubs in the citation network.

2.3 Temporal and Semantic Analysis

To explore how the network and the content of legal precedent evolved over time, we perform a temporal segmentation of the medical law sub-network aligned with South Korea’s political eras. For the temporal analysis, the dataset is partitioned into segments corresponding to distinct presidential administrations in South Korean history (from Syngman Rhee through Yoon Suk-yeol)¹. Within each period, we aggregate statistics such as the number of cases and distribution of centrality measures, to observe trends over time. For the semantic analysis, we examine the content of judgments utilizing AI and Large-language models (LLM). Specifically, for each administration we identify the cases in the medical law network with the highest popularity and authority and categorize each by case type (criminal, civil, administrative, and etc.). We then perform a qualitative content analysis of these key cases’ judgment text to extract

¹Presidents (and one’s administrations) with periods are — Syngman Rhee (1948–1960); Yun Posun (1960–1962); Park Chung Hee (1963–1979); Choi Kyu-hah (1979–1980); Chun Doo-hwan (1980–1988); Roh Tae-woo (1988–1993); Kim Young-sam (1993–1998); Kim Dae-jung (1998–2003); Roh Moo-hyun (2003–2008); Lee Myung-bak (2008–2013); Park Geun-hye (2013–2017); Moon Jae-in (2017–2022); Yoon Suk-yeol (2022–2025).

recurring themes and keywords that characterize each political era. This involve
identifying references to specific legal issues (e.g., unlicensed practice, malpractice,
patient consent, etc.) and noting how their prevalence change from one period to
another. By comparing dominant case types and recurring themes across different eras,
we identify shifts in the substantive focus of jurisprudence. The combination of
structural (network) analysis with qualitative (semantic) review enables us to link
changes in network structure to underlying shifts in legal doctrine and social context.

3 Results: Korean Legal System

3.1 Scale-Free Architecture

Dataset and Network Construction

The foundation of this analysis is a comprehensive dataset of 86,161 Supreme Court of
Korea cases decided between 1951 and 2023. From this corpus, a directed citation
network is constructed, resulting in 58,555 nodes ($|V|$ in our DG , cases with at least
one citation link) and 115,790 edges ($|E|$, citation relationships). After removing
isolated cases, the analysis distinguishes between the referring network and the
referred network. To measure influence and prestige, we focus on the referred network,
as in-degree serves as a direct proxy for a precedent's impact on future rulings [2].

Preferential Attachment

An analysis of the in-degree distribution of the referred network reveals a
non-Gaussian, heavy-tailed pattern. When plotted on a log-log scale, the tail of the
distribution exhibits an approximately linear relationship — a hallmark of a power-law
distribution and a characteristic feature of scale-free networks [11]. The probability
 $P(x)$ that a case is cited x times follows the form $x^{-\alpha}$. For the comprehensive legal
network, this exponent is estimated $\alpha_c = 3.29$ for the tail of the distribution. This
finding is consistent with studies of other citation networks, which often find
exponents in the range of 2 to 4 [12].

This scale-free structure is theorized to emerge from a generative mechanism known
as *preferential attachment* (or *cumulative advantage*). In this model, new nodes entering
the network have a higher probability of connecting to existing nodes that already have
a high degree. In the legal context, this translates to a *rich-get-richer* phenomenon: cases
that are already prominent and frequently cited are more likely to be discovered and

cited again by judges in future rulings, leading to the formation of highly influential hub precedents. The legal doctrine of *stare decisis* – the principle of adhering to precedent–can be understood as the fundamental behavioral algorithm that drives this network self-organization. It is not merely a legal norm but also a cognitive heuristic employed by judges to ensure stability, maintain legitimacy, and manage the complexity of decision-making [13]. When authoring an opinion, a judge is more likely to encounter, consider, and cite cases that are well-established within the legal canon. This judicial behavior is essentially a real-world implementation of the abstract preferential attachment model. The empirical discovery of a scale-free architecture in the Korean legal network thus provides compelling evidence that this micro-level cognitive rule shapes the macro-level structure of the entire judicial system. Such power-law patterns and the resulting inequalities are common in real-world complex networks across various domains [14,15].

Quantifying Inequality of Influence

The inherent inequality of influence within this scale-free structure can be further quantified using a Lorenz curve and the Gini coefficient G . This coefficient for the distribution of citations is calculated to be $G = 0.604$ for the referred network. The observed value confirms a high degree of inequality, indicating that a small fraction of legal precedents command a disproportionately large share of all citations, reinforcing the hub-and-spoke nature of the network.

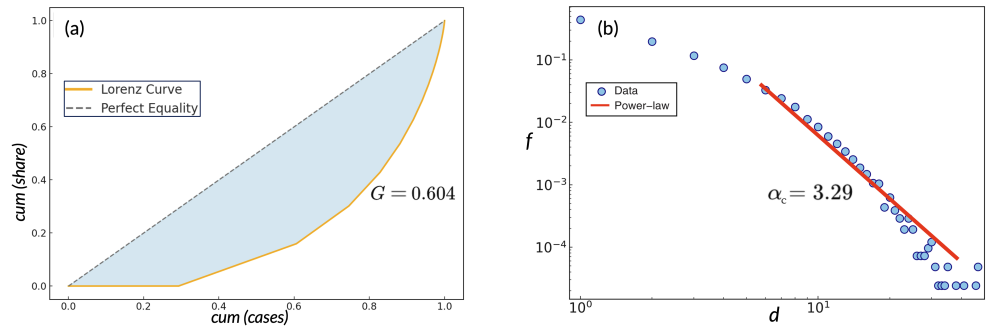


Fig 2. Scale-free structure and inequality in the referred citation network. (a) Lorenz curve (orange) versus the line of perfect equality (gray dashed). It summarizes inequality in the in-degree distribution, and the Gini coefficient 0.604. (b) Tail of the in-degree distribution on log-log axes with empirical frequencies (blue circles) and a fitted power-law over the tail (red line). Here d denotes in-degree and f denotes the normalized frequency. It yields an exponent $\alpha_c = 3.29$. Together, these panels indicate a heavy-tailed, scale-free architecture with influence concentrated in a small set of hub precedents.

Cognitive and Behavioral Framing

As a complementary interpretation, a network-based lens highlights how judicial cognition and behavior contribute to the emergent structure. *Stare decisis* functions as a behavioral heuristic that guides judges confronting complex issues to anchor on well-established authorities, reflecting bounded rationality and cognitive ease. In practice, judges often *follow the majority* by repeatedly citing canonical cases that peers cite, accelerating accumulation on already central nodes. This feedback loop between micro level citation choices and the macro level hub–spoke topology helps generate and reinforce the observed scale-free pattern. This heuristic choice pattern induces a preferential attachment dynamic, whereby new opinions disproportionately cite already central precedents, further entrenching network hubs.

3.2 Keystone Precedents: Popularity *vs.* Authority

While popularity provide a basic measure of a precedent’s impact, they do not tell the whole story. A case might be cited frequently for a narrow point of law –making it *popular*–yet those citations could come mostly from relatively minor cases. Another case might have fewer total citations but is referenced by several highly cited cases, embedding it in the core of doctrinal development. We distinguish these concepts by comparing popularity with authority as measures of influence in the network [16].

Authority analysis indeed reveals a different ranking of influential cases than simple popularity. The measure considers not just the quantity of citations but their quality (i.e., citations from authoritative cases count more). We compute centrality for all cases in the network and identify the top authority cases. Table 1 illustrates the contrast between the two metrics by listing exemplary precedents that rank highly by popularity *vs.* by authority. A clear pattern emerges: some cases with the highest popularity do not have the highest authority, and vice versa [2, 16].

- Case 95Da28625 (1997): This civil law case received 66 citations, the most of any case in the dataset, marking it as the single most popular precedent by raw citation. However, its authority is only the 10th highest in the network. In other words, although it is widely cited (perhaps for a broadly applicable rule or as a routine reference), it is not cited by the network’s most important hub cases to the same extent. It functions as a widely used legal tool, but not as a foundational pillar of doctrine.
- Case 82Do3248 (1983): This criminal law case has a more modest popularity with

Table 1. Top Authoritative Precedents by Popularity and Authority

Case Metadata				Judgment & Influence			
Case ID	Decision Date	President	Case Type	Judgment Type	Case Title	Popularity	Authority (1E-4)
82Do3248	1983.03.08	D. Jeon	Criminal	Judgment	Violation of the National Security Act, etc.	30	3.64
90DaKa23899	1991.12.24	T. Rho	Civil	En Banc Judgment	Insurance Claim	38	3.46
90Da12243	1991.12.24	T. Rho	Civil	En Banc Judgment	Land Ownership Transfer Registration	47	3.06
90Da8121	1990.12.11	T. Rho	Civil	Judgment	Ownership Transfer Registration	9	2.86
86DaKa1876	1987.05.26	D. Jeon	Civil	En Banc Judgment	Promissory Note	35	2.75
86DaKa83	1986.09.23	D. Jeon	Civil	Judgment	Revocation of Fraudulent Act	11	2.72
90Do357	1990.05.15	T. Rho	Criminal	Judgment	Violation of the Punishment of Violent Acts Act, etc.	33	2.64
91Da35571	1992.05.26	T. Rho	Civil	Judgment	Damages (Preliminary)	16	2.58
82Da3708	1983.07.12	D. Jeon	Civil	En Banc Judgment	Return of Unjust Enrichment	24	2.55
95Da28625	1997.08.21	Y. Kim	Civil	En Banc Judgment	Registration of Transfer of Ownership	66	2.20

30 citations, less than half that of Case 95Da28625. Yet it holds the #1 authority position in the entire network. This indicates that Case 82Do3248 is frequently cited by other highly influential cases. It sits at the center of a web of authoritative precedents, making it a critical axial case that undergirds a large portion of legal reasoning in the network. In effect, this case is a cornerstone of jurisprudence: many important rulings build upon it, even if not all lower-level cases cite it.

As an illustration of authority distinct from raw popularity, Fig. 3 shows the referred ego-network of 82Do3248, where numerous later cases converge on this precedent. This high authority case lies structurally central and exhibits inherited authority via citations from influential neighbors.

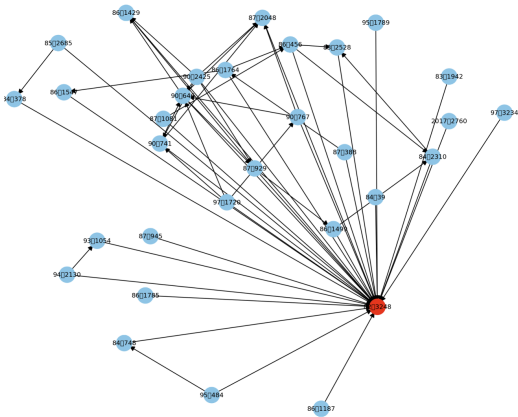


Fig 3. Ego-network of Case 82Do3248 in the referred citation network. Nodes represent cases and directed edges point from the citing case to the cited case. The focal precedent 82Do3248 (red) receives incoming citations from numerous cases, many of which are themselves influential, exemplifying authority that is not captured by raw citation counts alone.

These comparisons highlight that authority is not synonymous with popularity in the legal citation network [2,4]. High popularity cases may represent versatile and

commonly invoked rules, i.e., the building blocks of law, whereas high authority cases represent deep, principle-setting precedents that form the structural backbone of legal doctrine [17]. By identifying precedents that score high on authority, we pinpoint the jurisprudential keystones: decisions that have shaped the trajectory of legal reasoning in a fundamental way. This network-based insight provides a more nuanced, cognitive architecture of law, distinguishing surface-level popularity from embedded authority [2, 16].

4 Results: Medical Jurisprudence

4.1 Concentrated Hierarchy

To investigate how a specialized legal field organizes itself, the analysis now narrows its focus to the medical-related sub-network. This sub-network is constructed by filtering the comprehensive dataset for cases containing medical-related keywords in their titles or judgment statements, resulting in a focused network for domain-specific analysis.

The degree distribution of the referred medical network also exhibits a heavy-tailed, power-law characteristic, confirming that it too is a scale-free network. Focusing on the sub-network, we find that this domain exhibits an even more concentrated hierarchy than the legal system at large. The estimated power-law exponent for this one is $\alpha_m = 2.14$, and this is significantly lower than the exponent for the comprehensive legal network ($\alpha_c = 3.29$). A lower power-law exponent signifies a greater degree of inequality in the node connections. This indicates that the medical law network is even more dominated by a few central hubs than the legal system as a whole. The legal discourse within this specialized domain is more concentrated, with a smaller number of super-hub precedents commanding a vastly larger share of influence.

This increased structural hierarchy is not an anomaly but rather an emergent property of knowledge specialization itself. As a field of discourse becomes more technical and domain-specific, it tends to organize both cognitively and structurally around a smaller set of foundational, first-principle precedents. Experts in a specialized field rely on a core set of axioms from which subsequent reasoning flows, whereas generalists may draw from a wider, less concentrated pool of knowledge. The network's mathematical structure is a direct reflection of this cognitive organization. Specialization intensifies hierarchy, concentrating influence in a few core precedents. The lower alpha exponent serves as a quantitative signature that the *collective mind* of medical law is more focused and axiomatically structured than the broader legal

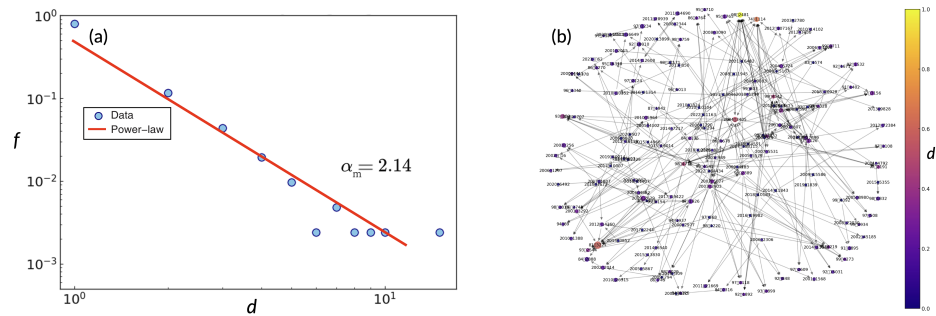


Fig 4. Heavy-tailed structure and concentrated hierarchy in the medical sub-network. (a) Tail of the referred degree distribution on log-log axes with empirical frequencies (blue circles) and a power-law fit over the tail (red line); d denotes in-degree and f the normalized frequency. It estimates $\alpha_m = 2.14$ (lower than the comprehensive network’s $\alpha_c = 3.29$). (b) Directed visualization of the medical-law sub-network. Here nodes are cases and arrows point from the citing case to the cited case; node color encodes normalized degree d (color bar). These panels indicate a more concentrated, hub-dominated hierarchy within this specialized domain.

4.2 The Political and Semantic Evolution of Medical Law

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To understand how the medical law network has evolved, the data is segmented by South Korean presidential administrations, from Syngman Rhee (1948–1960) to Yoon Suk-yeol (2022–2025) [18]. This approach connects the structural evolution of legal precedent to the nation’s broader political and social history, a period marked by authoritarian rule, democratization, and consolidated democracy. Using LLM and AI analysis, we capture meaning of the context in lawsuit cases.

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Analyzing the evolution of the medical law sub-network over time reveals that the influence of precedents and the content of legal discourse have undergone significant shifts aligned with South Korea’s political transitions.

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Structural Influence Over Political Eras

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In Fig 5, heatmaps visualizing the distribution of authority over time reveal distinct periods of precedent formation and influence.

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A notable spike in authority appears during the Park Chung-hee era (1963–1979), an authoritarian period, indicating the early formation of structurally dominant medical cases that established foundational, state-centric regulations. In fact, a notable cluster of high-authority cases originates in the late 1960s and 1970s [19]. These cases established foundational regulations (often state-power oriented) for the medical field. The strong

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authority of these early precedents suggests that even under authoritarian rule, some landmark decisions set the groundwork that would persist for decades.

In contrast, the Kim Young-sam administration (1993–1998), which marked South Korea’s transition to a full-fledged democracy, shows a more consistent distribution of authority across multiple rulings [20,21]. Instead of one or two cases dominating, we see multiple cases each contributing moderately high authority. This reflects a diversifying of legal discourse as the judiciary in a newly democratic society grappled with a broader array of issues and exercised greater independence. The legal framework for medical disputes is being built out in multiple directions (e.g., professional regulation, patient rights, insurance, etc.) rather than dictated by a few top-down rules.

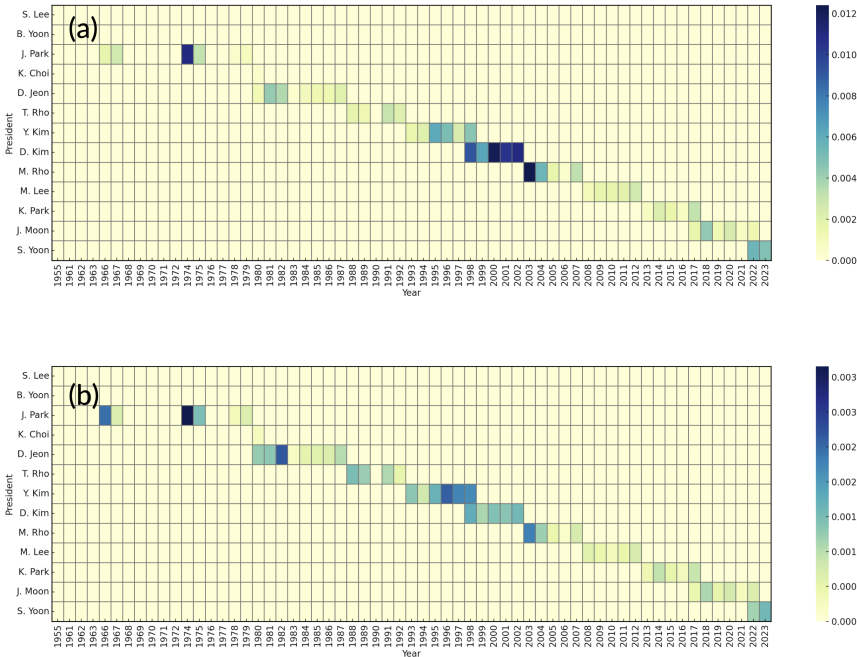


Fig 5. Temporal heatmaps of citation activity and authority in the medical sub-network (by presidency and year). (a) Heatmap of citation activity in medical cases aggregated by calendar year and presidential administration; darker cells (as in color bar) indicate years in which medical cases attracted more citations. (b) Heatmap of authority likewise aggregated by year and presidency; darker cells mark periods when highly influential medical precedents emerged or were reinforced. Rows are presidential eras (initialed romanization as in the footnote 1) and columns are years.

Subsequent eras, such as the late 1990s and 2000s under Kim Dae-jung (1998–2003) and Roh Moo-hyun (2003–2008), show spikes in citation activity — many new medical cases were decided and heavily cited during these times (coinciding with rapid social and institutional changes). These spikes indicate periods of expansion and

consolidation of medical jurisprudence. The courts were actively working out new legal questions in healthcare, and referencing established precedents intensively to build consistent doctrine. By the 2010s such as the Moon Jae-in era (2017–2022), the network structure appears robust and mature, with influence dispersed across a network of interlinked precedents, which is a reflection of a stable, institutionalized legal domain.

Semantic Evolution: from Crime to Care

The most significant finding from the temporal analysis is a profound qualitative shift in the nature of the most influential medical cases over time. This *semantic shift* is identified through an analysis of the case types (criminal, civil, administrative) and keywords from the judgment statements of hub cases in each presidential era.

- **Early Periods (Authoritarian/Developing State):** During the administrations of Park Chung-hee and Chun Doo-hwan, the most central medical cases are predominantly criminal in nature. Key themes revolved around the “Violation of Medical Law,” such as unlicensed medical practice. This reflects a state-centric focus on establishing basic professional standards and regulating the medical profession through punitive measures, a common feature in the early development of a national medical system [18].
- **Later Periods (Democratization/Mature State):** Beginning in the 1990s and accelerating through the 2000s and 2010s, the focus of influential precedents shifts dramatically toward civil and administrative law. Keywords associated with hub cases in the eras of Kim Dae-jung, Roh Moo-hyun, and Moon Jae-in include “Duty of care, Damage compensation, Administrative disposition, Negligence”. This change reflects a societal evolution toward emphasizing patient rights, medical accountability, and the complex governance of a modern healthcare system. The legal discourse moves from punishing rogue practitioners to adjudicating nuanced disputes over the quality of care and the rights and responsibilities of both patients and providers [22].

Table 2 shows this thematic transformation, showing the dominant case types and themes for illustrative eras. In the early period, medical jurisprudence served the state’s interest in establishing order (penalizing misconduct by medical practitioners). In the later period, it increasingly served the citizens’ interest in protecting rights, which include adjudicating claims of malpractice, enforcing standards of care, and managing the healthcare system’s administrative fairness.

Table 2. Thematic Shifts in Medical Jurisprudence Across Presidential Eras

Presidential Era	Dominant Case Type(s)	Key Semantic Themes / Keywords	High-Authority Case (e.g.)
Early Authoritarian (J. Park / D. Chun)	Criminal	Policing the profession; State regulation; Violation of Medical Law	74Do1114, 81Do3227
Democratic Transition (T. Roh / Y. Kim)	Criminal, Administrative	Professional standards; Licensing; Interpretation of medical law; Administrative disposition	92Nu16010, 97Do1234
Consolidated Democracy (D. Kim / M. Roh / Moon)	Civil, Administrative	Patient rights; Accountability; Duty of care; Damage compensation; Negligence	2004Do3405, 2017Do17699

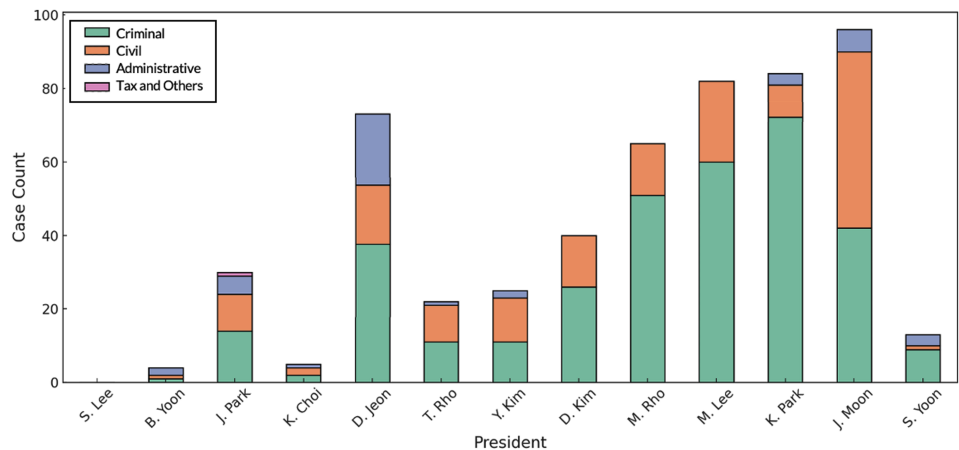


Fig 6. Semantic shift in influential medical cases across presidential eras. Stacked bars display the distribution of case types in the medical sub-network by presidency (Criminal, Civil, Administrative, Others). Early authoritarian administrations are dominated by criminal rulings, whereas later democratic eras show increasing shares of civil and administrative disputes — indicating a move to adjudicating duties of care, compensation, and administrative governance. Axis labels use initialed romanization of presidents (See the footnote 1).

This temporal-semantic analysis provides empirical evidence that the law has mirrored South Korea’s broader socio-political evolution. The shift from criminality to care in influential cases reflects how legal priorities reoriented to vindicating individual rights and addressing complex civil governance of healthcare. In effect, the citation network encodes a record of social perception and values: as public consciousness around healthcare changed such as greater awareness of patient rights, higher expectations of medical accountability, the key legal precedents in that domain changed accordingly [22,23]. The structure and content of the network thus serve as a high-fidelity sensor of societal behavior change, capturing in medical law cases [18,22].

5 Discussion: Legal System as Collective Intelligence

Synthesizing the empirical findings, our analysis supports a new theoretical lens that views the judicial system as a form of *collective intelligence*. The legal citation network is

not merely a static archive of decisions, but a dynamic, evolving system that exhibits memory, logic, and adaptation. The network is the concrete expression of a shared group intelligence emerging from the collaboration and cumulative efforts of many individual judicial actors. Each citation represents an act of information retrieval and reinforcement. The aggregation of millions of such individual decisions produces a complex system that learns and adapts in response to external environmental changes (i.e., social and political shifts), all without a central designer or controller. The principle of precedent functions as the system’s memory, the backbone of high-authority cases forms its structural core of reasoning, and the observed semantic shifts demonstrate its capacity for adaptation.

The Emergent Mind of the Law

Complex adaptive systems theory suggests that when many agents interact over time, higher-level order and intelligence can emerge [24]. Our results illustrate this in the legal context. The citation network of precedents functions as an emergent legal mind that exhibits properties of memory, learning, and adaptation. The principle of *stare decisis* provides memory — past decisions are preserved and continuously cited, allowing the system to remember and build on prior knowledge. The network’s structure, especially the backbone formed by high-authority cases, provides a logical architecture of legal reasoning, akin to the hardwiring of a brain that channels signals along certain dominant pathways. We also observed the system’s capacity to adapt: as external societal conditions changed (authoritarian to democratic, paternalistic to rights-oriented healthcare), the network’s content and structure shifted accordingly, demonstrating a form of collective learning. No single judge or court expressly decided to reorient medical jurisprudence from criminal law to civil rights, yet through many individual decisions the system as a whole achieved that transformation. This self-organizing adaptation is a hallmark of a collective mind in action.

Cognitive Heuristics and Judicial Behavior

The macroscopic structure of the legal network is deeply linked to the microscopic foundations of human cognition. The process that generates a scale-free structure, the preferential attachment, is driven by the cognitive heuristics judges employ to manage overwhelming complexity. Faced with a vast corpus of prior cases, judges naturally rely on mental shortcuts. A *follow-the-majority* heuristic (the tendency to imitate the

citations of other judges) is especially influential [25]. This heuristic, combined with the greater availability and perceived legitimacy of already well-cited precedents, translates directly into the *rich-get-richer* dynamics observed.

The emergent structural patterns of the legal network are partly a product of bounded rationality and intuitive thinking by its human participants [26]. Empirical psychological studies have documented various cognitive biases and decision-making shortcuts in legal contexts (for jurors and judges alike) [27,28], and our findings suggest these biases cumulatively manifest in the large-scale topology of the law. Concretely, micro level rules —“cite what others cite,” availability, and perceived legitimacy— aggregate into macro-level regularities: heavy-tailed distributions, hub formation, and layered (deep *vs.* shallow) influence strata across precedents; this is the explicit bridge from judicial cognition to network architecture.

Social Perception Mirrored in Network Dynamics

The semantic evolution witnessed in the medical law sub-network provides a compelling example of how the legal system acts as a high-fidelity sensor of societal consciousness. The shift in dominant themes from state control to individual rights in medical jurisprudence was not an isolated legal phenomenon; it encoded and institutionalized a profound change in social values concerning healthcare, professional authority, and personal autonomy. In other words, the structure and content of the legal network serve as a durable, data-driven archive of social evolution. Law, through its network of precedents, captures how societal norms influence judicial outcomes and, reciprocally, how judicial decisions influence society’s expectations and behavior. This perspective resonates with interdisciplinary scholarship highlighting the reciprocal influence of law and social values [29]. It underscores that legal developments can be better understood as both products and drivers of broader social consciousness.

Implications for Policy and Artificial Intelligence in Law

Our analysis carries important substantive implications. For legal practitioners and scholars, distinguishing between precedents that are merely popular and those that are truly authoritative can inform strategic litigation choices, legal education, and reform efforts. Knowing which hub cases underpin the structure of doctrine can help predict which precedents, if altered or overturned, would have the widest ripple effects on the legal system [27]. For policymakers, visualizing the evolution of legal discourse

provides a tool to assess the impact of past legislative or institutional changes and to identify domains where legal doctrine may be lagging behind societal norms.

Moreover, given the rapid rise of artificial intelligence in the legal field, a structural understanding of jurisprudence is more critical than ever. AI systems trained on legal corpora (such as large language models) will inevitably learn and reproduce patterns inherent in the data including the network’s hubs, clusters, and potential biases [30]. A thorough comprehension of these emergent structures is essential to ensure that such tools are developed and deployed responsibly, enhancing rather than distorting the administration of justice [31, 32].

In practice, ranking and retrieval pipelines should weight authority above raw popularity, run popularity-bias audits, and surface cross-community diversity to avoid over-reliance on routine but shallow precedents. Conversely, when exploration is desirable, systems can up-weight novel yet authority-adjacent cases to balance exploitation with exploration, aligning machine assistance with expert juristic reasoning. As initiatives on AI and access to justice emphasize, technology should be harnessed to reinforce the rule of law and expand legal access, not to inadvertently amplify existing inequities in the legal system.

6 Conclusion

We demonstrate that the citation network of South Korea’s Supreme Court decisions is a complex, scale-free system generated by the aggregate behavior of individual judges over seven decades. A small subset of precedents has emerged as the network’s structural backbone, reflecting a process of cumulative advantage reinforced by the practice of *stare decisis*. Within the specialized domain of medical law, the network exhibits an even more pronounced hierarchical structure — a quantitative signature of how concentrated expert knowledge can become. Using dual metrics we distinguish between precedents that are widely cited and those that are deeply foundational, uncovering the keystone cases that form the basis of legal reasoning in this system. Critically, our research also maps a clear structural and semantic evolution in the network, revealing how the focus of high-impact jurisprudence shifted from state control to individual rights as South Korea transitioned from authoritarianism to democracy.

Viewing the legal system through the lenses of *perception, cognition, behavior, mind, and intelligence* provides a more holistic understanding of jurisprudence. The network

of precedents operates as a distributed cognitive system. It is a form of collective legal mind that records societal memory, uses heuristics to manage complexity, and adapts to new information and values. This perspective opens new avenues for interdisciplinary research, bridging law with data science and AI, cognitive psychology, and complexity theory. In practice, network insights support targeted legal education on seminal cases, bias-aware legal reasoning that privileges authority over popularity, and responsive lawmaking aligned with evolving social norms.

In summary, this study illustrates the power of integrating structural and semantic analysis to narrate the story of a legal system's evolution. Our findings not only contribute a detailed empirical account of legal history but also exemplify a methodology applicable to other jurisdictions. As legal systems worldwide face rapid change and increasing complexity, such quantitative mappings of "legal genomes" will be invaluable for guiding the law's development in line with the collective intelligence and values of society.

Acknowledgments

Conflicts of Interest: The authors declare that they have no competing interests.

Data Availability: Through the law firm-affiliated commercial database system, de-privatized data is obtained. Access to the licensed de-privatized part is available under a data use agreement.

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Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination

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Abstract

Humans can intuitively associate phonological forms with visual properties, exemplified by sound symbolism effects such as “*bouba-kiki*,” where people consistently match certain pseudowords with particular shapes. In contrast, AI large language models (LLMs) like GPT-4 learn from text and may rely on learned semantic and distributional regularities rather than perceptual intuition. We compare humans and an LLM in a shape imagination task using non-symbolic stimuli: six novel *nonwords* (Nonword A–F). Forty multilingual participants ($n = 40$) read each nonword, imagined its pronunciation, and depicted a simple shape, while GPT-4 received the same strings as text input and generated a single shape description per stimulus. Human responses showed consensus consistent with intuitive sound-symbolic mapping. GPT-4, by contrast, produced shapes that diverged from the human majority, suggesting a reliance on language-based associations rather than pre-linguistic crossmodal intuition. The resulting mismatch highlights a cognitive gap between human intuitive processing and the LLM’s semantic reasoning, and provides empirical evidence relevant to the symbol grounding problem. Our findings motivate future work on perceptual grounding and multimodal learning in foundation models.

Keywords: sound symbolism, nonwords, symbol grounding, multimodal cognition, AI, large language models, intuition

1. Introduction

Language is traditionally viewed as an arbitrary symbolic system, yet humans demonstrate remarkable intuitive linkages between sensory modal-

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ities and linguistic sounds. A classic example is the “*bouba-kiki*” effect, in which people overwhelmingly associate the pseudoword “*kiki*” with a spiky shape and “*bouba*” with a rounded shape. This cross-modal mapping occurs across different age groups and cultures, suggesting human cognition. Such sound symbolism indicates that human perception can operate on a pre-linguistic intuitive level: certain sounds inherently *feel* round or sharp, guiding visual imaginings of shape. These intuitions are thought to arise from sensory-affective cues-properties like the sharpness or softness of a sound evoke corresponding visual textures or forms.

Large language models (LLMs), by contrast, acquire knowledge from textual data. AI models like GPT-4 have demonstrated impressive language understanding and generation capabilities (e.g., in storytelling, question-answering, and coding tasks) by learning statistical patterns in massive corpora. However, they lack direct embodiment or sensory inputs; their *experience* of the world is mediated entirely through words. This raises a pertinent question: can an LLM, armed with semantic associations but devoid of direct perception, emulate the intuitive sound-shape mapping that humans perform? Or will it rely on fundamentally different (perhaps more literal or culturally learned) mechanisms to *imagine/visualize* a sound?

This question ties into the broader issue of *symbol grounding*. Humans ground linguistic symbols in sensorimotor experience — for example, the word “smooth” relates to tactile and visual sensations of smoothness. In an LLM, symbols are grounded only in other symbols (textual contexts), leading to what Harnad (1990) termed the “*symbol grounding problem*”. Without grounding in real-world perception, an AI’s understanding of terms like

“smooth sound” or “jagged noise” may differ qualitatively from a human’s. The LLM might draw upon semantic memory (e.g., phrases it has seen that relate sound to shape) rather than any innate cross-sensory intuition.

To investigate these issues, we designed an experiment comparing human and AI (GPT-4) performance on a phonological visualization task using written stimuli. Human participants were presented with six novel nonwords; they read each word, imagined its pronunciation, and depicted it as a simple visual shape. Separately, GPT-4 received the same nonword strings as text input and was prompted to generate a shape description.

By analyzing the correspondences (or lack thereof) between the shapes produced by humans and the AI, we aim to uncover differences in cognitive strategy: do humans rely on affective, intuitive mappings while the LLM leverages language-based reasoning? We hypothesized that human participants would show systematic and intuitive nonword-to-shape mappings (consistent with cross-modal intuition seen in sound symbolism literature), whereas GPT-4’s outputs would diverge due to its reliance on semantic knowledge without direct sensory grounding.

Our study contributes to understanding how modern AI’s generative mechanisms differ from human cognition in a multimodal context. Demonstrating a gap between human intuition and an LLM’s semantic reasoning would highlight the need for incorporating perceptual grounding into AI models. Conversely, if an LLM were to surprisingly align with human responses, it could suggest that some aspects of cross-modal mapping are learnable from language data alone.

In the following, we describe our experimental methods, present re-

sults contrasting human and GPT-4 generated visualizations, and discuss the implications for cognitive science and AI development.

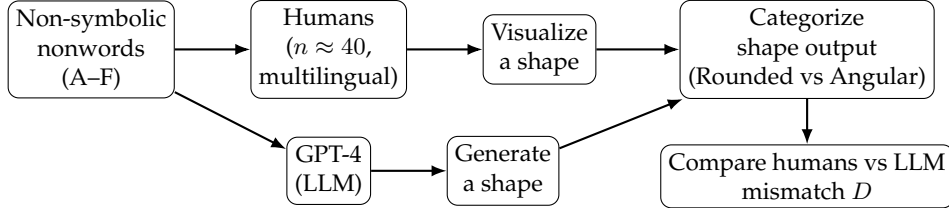


Figure 1: Overview of the experimental workflow. Humans read/pronounce (mentally) nonwords and imagine pronunciation to visualize shapes, while GPT-4 processes the same strings as text input. Outputs are categorized as *Rounded* or *Angular*. Results are analyzed using a mismatch score D .

2. Methods

2.1. Stimulus representations for humans and the LLM

Human participants received the stimuli as short written nonwords (A–F). The IPA (International Phonetic Alphabet) transcriptions are provided in this manuscript only to document the intended pronunciations and to label stimuli consistently; for humans, the IPA was used only as a pronunciation reference. In contrast, GPT-4 could not process this as an audio or sound in our setup. We therefore provided GPT-4 with orthographic *English-like spellings* that approximate the target pronunciations, and asked it to generate a corresponding basic shape from the written non-word prompt. Table 1 summarizes the stimulus set and the representations used for human participants and the LLM.

Table 1: Stimulus labels, IPA transcriptions (documentation), and orthographic spellings. Spellings are constructed to approximate the intended pronunciations.

Label	IPA	English spelling
A	/ˈʌn.guː.i.gʌn/	Ungooigan
B	/ˈæŋ.gəl.zər/	Angulzer
C	/ˈkjʌn.bʌk.kuː.neɪ/	Kyunbokkunay
D	/ˈpou.kə.ɪə/	Pokera
E	/ˈpaː.tuː.miː/	Patoomi
F	/ˈgæl.uː.piː/	Galupee

2.2. Participants

A total of forty participants ($n = 40$) were recruited for the human experiment. Participants were young adults of diverse linguistic backgrounds, including native speakers of Korean, Chinese, English, and French. All were proficient in English, and instructions were provided in English. The sample included roughly equal numbers of male and female participants, with ages ranging from late teens to middle adulthood. No participant reported any form of synesthesia or reading or visual impairment. Participation was voluntary in accordance with institutional guidelines.

2.3. Stimuli and Task

The stimuli consisted of six novel *nonwords*, labeled Nonword A through Nonword F. Each stimulus was a short written string constructed to be non-symbolic (i.e., not a real word in the participants’ languages) while retaining clear phonological structure (plausible consonant–vowel patterns and syllable shapes). The set was designed to span a range of phonological impressions when read and internally pronounced (e.g., strings that tend to feel “smooth” versus “sharp” in their imagined articulation). For brevity,

we refer to these non-symbolic words as *nonwords*.

Stimulus	Non-symbolic sound word	TASK Humans: read/pronounce (mentally) and sketch a simple shape LLM (GPT): given the same string, generate a shape Categorize: <i>Rounders</i> vs. <i>Angular</i>
A	'ʌn.gu:.i.gʌn	
B	'æŋ.gəl.zər	
C	'kʃʌn.bək.ku:.nei	
D	'pʊθ.kə.rə	
E	'pɑ:.tu:.mi:	
F	'gæl.u:.pi:	

Figure 2: Stimulus set and task. Six nonwords were presented as stimuli. Human participants were asked to read each string (self-generated auditory imagery), and they imagined its pronunciation to visualize a shape. GPT-4 received the same strings as text input.

Participants were tested individually in a quiet room. They were instructed to *read/pronounce (mentally)* each nonword presented and then “visualize the nonword” by imagining a single shape or form that they felt corresponded to the *imagined pronunciation* of the string. Importantly, no example shapes were provided to avoid biasing responses. After reading (and silently self-pronouncing), participants either sketched a simple shape on paper or spoke that shape. In practice, most participants opted to draw or annotate a shape that represented their intuitive impression. They were encouraged to keep these visualizations simple and to focus on the overall form (*Rounded vs. Angular*), rather than adding extraneous details. The task was self-paced, with participants proceeding to the next stimulus after recording a visualization for the current one. The order of presenting the six nonwords was randomized for each participant to counterbalance any order effects.

2.4. LLM (GPT-4) Procedure

To generate AI-based visualizations for comparison, we used GPT-4 (an LLM by OpenAI). GPT-4 was given the *same written nonwords strings* (Nonword A–F) as text input. For each stimulus, the model was instructed with a consistent prompt, for example: “You are given a novel, meaningless nonwords provided as English-like spellings. Express how it would sound if pronounced, then describe a single simple visual shape (*Rounded* or *Angular*) that best matches that imagined sound.” This setup ensures that both humans and the model operate over the same written representation of the stimulus, while differing in the mechanisms used to derive a shape internally.

The model was run with a temperature setting favoring consistency to minimize randomness in outputs. Using a fixed prompt and fixed generation settings, we queried GPT-4 once per nonword and used the resulting response as the primary description. In general, GPT-4 complied by providing a short paragraph or sentence describing a geometric or symbolic shape for each stimulus; these descriptions often included analogies or similes characterizing the envisioned form.

The GPT-4 outputs were interpreted by the experimenter into the same shape categories as the human expressions. For example, if GPT-4 described “a jagged, star-like shape with sharp edges,” this was coded as an *Angular* shape. If GPT-4 described “a smooth orb or blob,” it was coded as *Rounded*. In cases where GPT-4’s description combined multiple elements, the predominant global shape quality was used for classification.

2.5. Data Analysis

For each of the six nonword stimuli, we collected the form of shapes (human) and the single GPT-4 generated shape description (AI). The primary analysis focused on categorizing each response as either *Rounded* or *Angular*, reflecting the two fundamental shape classes of interest. We defined *Rounded* shapes as those composed of curved lines or circular forms with no sharp corners (e.g., circles, ovals, wavy blobs). *Angular* shapes were defined by presence of pointed corners or straight-line edges (e.g., triangles, zig-zags, star shapes). This binary classification was motivated by the theoretical relevance of round *vs.* sharp distinctions in sound symbolism (as in *bouba vs. kiki* shapes) and by the nature of our stimuli which were expected to evoke either smoother or sharper impressions.

The GPT-4 output for each sound was similarly classified. We then quantified, for each sound, the proportion of human participants who chose a *Rounded* shape and the proportion who chose an *Angular* shape. These were compared to the category of the GPT-4 output for that sound.

To assess overall concordance or discrepancy between human and AI responses, we suggested a mismatch score (section 2.6) which incorporates how often GPT-4’s chosen shape category matched the majority human response for a given sound. That also aggregated results to see the overall distribution of shape types chosen by humans across all sounds *vs.* by GPT-4. Because GPT-4 provides only one data point per stimulus, formal statistical comparison is not straightforward; however, we used descriptive and inferential approaches with caution by utilizing D . We treated the human responses as observed frequencies and GPT-4’s single response as an ex-

pected outcome. A qualitative and heuristic comparative analysis was also appropriate given the data, focusing on patterns of divergence.

2.6. Quantifying Human–LLM Mismatch

To summarize the divergence between human and LLM responses, we define a simple mismatch score D , that compares the human response distribution to the LLM’s binary output.

Let $H_i \in [0, 1]$ denote the proportion of human participants who categorized nonword i as *Rounded*, and let $G_i \in \{0, 1\}$ denote the LLM’s binary judgment for the same stimulus ($1 = \textit{Rounded}$, $0 = \textit{Angular}$). Here, i means the given nonword. We then define the average absolute mismatch across N stimuli as:

$$D = \frac{1}{N} \sum_{i=1}^N |H_i - G_i|. \quad (1)$$

This metric captures how far the LLM’s discrete decisions deviate from the human consensus distribution. A value of $D = 0$ indicates perfect alignment between the LLM and human majority judgments, whereas larger values indicate greater divergence.

3. Results

3.1. Human Intuitive Shape Imaginations

Figure 3 illustrates a clear contrast between distributed human judgments and the LLM’s deterministic outputs.

Human participants exhibited clear and systematic mappings from non-word stimuli to imagined visual shapes. For each of the six, a strong majority of participants converged on the same basic shape category. For example, one of the stimuli (Nonword A, the low phonologically smooth non-word) was visualized as a *Rounded* form by 36 out of 40 participants (90%), often described or drawn as a gently curving blob or concentric circles. In contrast, another stimulus (Nonword E, the high phonologically sharp nonword) elicited predominantly *Angular* representations: 37 out of 40 participants (92.5%) depicted it with jagged, spiky forms (such as zig-zag lines or starburst-like shapes). The remaining sounds showed similarly polarized results. In total, four of the six sounds were predominantly mapped to one category (*Rounded* or *Angular*) by more than approximately 75% of participants. The remaining two sounds showed a slightly more mixed response, but even in those, a simple majority ($\geq 65\%$) favored one shape type. These findings confirm that participants were not guessing randomly; rather, they were leveraging a consistent intuitive sense of which shapes *fit* a given sound’s character.

Notably, this pattern held across participants. This aligns with the idea that the mappings observed are driven by human phonological–visual intuition, not by specific language semantics.

It echoes previous research showing that sound-shape correspondences like the “*bouba-kiki*” effect are robust. In post-experiment informal interviews, participants often reported relying on how the sound “felt” (e.g., “that sound just felt sharp to me” or “the sound seemed to have a circular motion”). This provides qualitative evidence that their decisions were

based on immediate sensory-affective impressions, which is an intuition, rather than logical reasoning or learned associations.

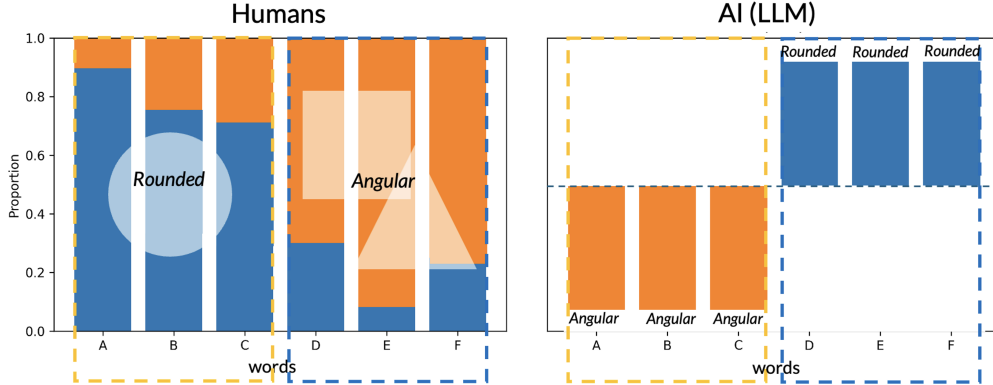


Figure 3: Human *vs.* LLM shape categorizations. Left: proportions of human responses categorized as *Rounded* or *Angular* for each nonword (A–F). Right: binary outputs produced by the LLM for the same stimuli. The average mismatch score across stimuli *D* is 0.766.

3.2. AI Generated Shape Descriptions

GPT-4 produced a single shape description for each nonword stimulus after processing the textual prompt. These descriptions were generally lucid and creative, indicating that the model attempted to make sense of the sounds in a visual way using its prior knowledge. However, the nature of GPT-4’s outputs was notably different from human responses. Crucially, in most cases the shape category described by GPT-4 did not match the dominant category chosen by humans for the same sound.

We also report brief illustrative excerpts of GPT-4 outputs (Table 2).

GPT-4’s envisioned shape was of the opposite type to the human majority response. For instance, for Nonword A (the phonologically smooth nonword which humans overwhelmingly mapped to a rounded shape), GPT-

Table 2: Illustrative GPT-4 shape descriptions (short excerpts) for two stimuli, highlighting meaning- or metaphor-driven associations.

Stimulus	GPT-4 output excerpt
Angulzer (B)	“a sharp, jagged triangle-like form”
Pokera (D)	“a smooth, rounded bubble-like shape”

4 described something akin to “a pointed spiral gradually unwinding,” which we classified as *Angular*. Conversely, for Nonword B (the phonologically sharp nonword that humans saw as jagged and angular), GPT-4’s description was along the lines of “several bubbles popping in succession, leaving round ripple shapes,” an imagery we coded as *Rounded*. In these cases, GPT-4’s output was essentially the inverse of the human intuitive mapping.

Across all six sounds, GPT-4’s output did not show a consistent bias towards always producing rounded or always angular shapes; instead, the divergence from human responses suggests that GPT-4 was leveraging a different basis for its decisions. The model might have been influenced by linguistic or cultural associations present in its training data. For example, GPT-4 may have *read* during training that certain types of sounds (like gentle hums) are metaphorically described as spirals or that popping sounds are associated with bubbles. In essence, GPT-4’s visualizations appeared to be *semantic* or metaphorical in nature-grounded in language and known concepts-rather than purely *perceptual* analogues of the sound.

3.3. Comparison Between Human and LLM

Figure 3 summarizes the comparison of human vs. GPT-4 shape categorizations. Humans overwhelmingly relied on what appears to be an af-

fective, intuitive mapping from phonological impressions to visual form: smooth-sounding nonwords (as internally pronounced) yielded soft, rounded visuals, while sharp-sounding nonwords (as internally pronounced) yielded sharp, angular visuals.

Using the six stimuli in our experiment ($N = 6$), we compute the mismatch score following equation (1). We obtain $D = 0.766$, indicating substantial divergence between the LLM’s binary judgments and the human response distribution.

GPT-4, lacking perceptual grounding, instead seemed to rely on semantic reasoning, producing visuals that sometimes conflict with those intuitive mappings. The systematic nature of the divergence—especially in cases where GPT-4’s choice was effectively the opposite of the human consensus—indicates that GPT-4 does not inherently possess the human-like crossmodal correspondences that might be expected if it had analogous perceptual grounding.

AI’s outputs relative to human intuition underscores a systematic gap. GPT-4 often chose a different category than humans did. Given the single-response nature of the LLM data, we focus on descriptive comparisons between human distributions and LLM outputs. Results of D (0.766) showed clear divergence beyond typical human variability, reinforcing that GPT-4’s output was not within the normal variability of human responses.

These results support our initial hypothesis: human cognition and the LLM’s generative mechanism operate differently in this multimodal visualization task. Human participants, even without any explicit instructions on how to perform the mapping, showed a high level of agreement, consistent

with a shared cognitive bias or intuition. GPT-4, given only textual descriptions, performed the task in a way that might be internally consistent (i.e., consistent with some learned language patterns) but is misaligned with human intuitions. The next section delves into the implications of these findings, discussing why these differences arise and how they relate to theories of cognition and AI design.

4. Discussion

Our findings reveal a pronounced contrast between human and AI (GPT-4) approaches to mapping imagined pronunciations of nonwords into visual forms, shedding light on deeper differences in cognitive processing. The human participants’ consistent sound-shape associations underscore the role of immediate, pre-linguistic intuition in cross-modal perception. In essence, humans seem to tap into an affective, embodied mode of cognition when faced with this task: the phonological impressions of the stimulus (pitch, smoothness, rhythm) directly evoke a certain felt shape *qualia* (roundedness or angularity) without the mediation of explicit language. This aligns with theories in cognitive science that propose much of human perception and analogy-making is grounded in our bodily and sensory experiences. Within our linguistically diverse sample, we observed broadly convergent mappings, consistent with the possibility that the effect reflects a shared human tendency rather than being driven solely by any single language background.

In contrast, GPT-4’s performance highlights the limitations of a system that, however sophisticated in linguistic pattern recognition, lacks em-

bodied grounding. GPT-4’s “knowledge” of sounds and shapes exists only through correlations it has encountered in text. Thus, when asked to imagine a shape for a sound, GPT-4 appears to rely on semantic associations—perhaps recalling descriptions from literature (e.g., “a piercing whistle like a needle”) or common metaphors (e.g., “booming sounds are round”). Its outputs often read as analogies that a person might produce if they were explicitly reasoning or being poetic, rather than the kind of nonverbal, gut-level mapping seen in the human drawings. This difference can be framed in terms of dual processing: the humans employed a fast, automatic, and intuitive mode of thinking, whereas GPT-4, in essence, functioned leveraging learned knowledge and explicit reasoning through language.

The results exemplify the *symbol grounding problem* in a concrete way. Harnad (1990) described the challenge for AI systems: how can symbols acquire meaning *intrinsically* rather than just parasitic on other symbols? Our experiment’s outcome suggests that GPT-4’s internal symbols for *sound* and *shape* are not grounded in the same way a human’s concepts are. A human reading a jagged-sounding nonword and imagining its pronunciation might literally experience an uneasy sensation that translates to a spiky image—an embodied semantic link. GPT-4, given a written nonword string, does not feel anything; it combs its training data for likely responses. The consequences are evident: the AI’s response, while sometimes superficially plausible, does not consistently align with the human intuitive response because the meanings of “jagged” or “smooth” for GPT-4 are ultimately tethered only to other words, not to sensory reality. In other words, GPT-4 can mimic the form of understanding (by producing relevant-sounding de-

scriptions) without sharing the actual grounded experience that underpins human understanding.

It is illuminating to consider the cases where GPT-4’s output did match the human consensus. How was the model able to *guess* correctly in those instances? Possibly, certain sound descriptions triggered well-known metaphors in text that happen to coincide with human intuitions. For example, a *phonologically smooth nonword* sound might have frequently been described in books as a “calm, wave-like” sensation, leading GPT-4 to produce a rounded shape description. This could indicate that for some very common correspondences, language corpora do capture a shadow of human embodiment (since humans, when writing, may encode their perceptual intuitions into text). However, this is an indirect and hit-or-miss route. The majority of our stimuli and responses show that where the correspondence is not something that appears often in prose or common expressions, GPT-4’s guesses can be off the mark relative to human perception.

Our study sits at the intersection of cognitive science and AI, and the implications extend to both fields. For cognitive science, the strong agreement among humans highlights the robustness of cross-modal correspondences in the absence of language. This relates to ongoing research in sound symbolism and synesthesia. The *bouba-kiki* effect is one famous example, but our results generalize that principle to a set of novel sounds, reinforcing theories that propose intrinsic connections between acoustic properties and perceptual shape qualities. Some researchers (Ramachandran & Hubbard, 2001) have speculated that such cross-modal mappings might have played a role in the origin of language. While our experiment did not directly ad-

dress language origin, the ease with which our participants performed the task suggests that translating sounds into images is a natural ability, possibly hinting at why certain sounds (phonemes) have non-arbitrary associations with meaning.

For AI research, our findings provide a clear demonstration of where current LLMs fall short compared to human cognition. GPT-4 is one of the advanced language models, even exhibiting emergent abilities that some argue approach aspects of general intelligence. Yet, here we see a domain-basic sensory intuition-where the model’s lack of direct experience creates a gap. This gap is not due to lack of capacity or knowledge per se (GPT-4 has *read* about shapes and sounds), but due to the qualitative nature of its knowledge. Our results resonate with arguments by AI scientists and philosophers that without grounding in the real world, AI’s understanding remains shallow or misaligned. As Bubeck et al. (2023) discuss, strong performance on many tasks can coexist with failures that reflect reliance on surface-level correlations rather than grounded understanding in certain domains.

One promising direction to address this is the development of *multi-modal* models. If an AI were trained not just on text but also on audio-visual data (sounds or videos) paired with language, it could begin to form the kind of cross-modal representations that humans do. Integrating such capabilities with language models could allow AI to ground its semantic predictions in sensory modalities. Our study provides empirical motivation for these efforts, demonstrating the current model’s shortcomings that might be overcome with grounded learning.

Another implication is the value of interdisciplinary evaluation of AI. By borrowing a paradigm from cognitive psychology (sound-shape matching) to test an AI, we gained insight into the AI’s mode of reasoning. This kind of evaluation goes beyond standard benchmarks of accuracy or perplexity and examines the *nature* of AI cognition in comparison to humans. It exemplifies how cognitive science can inform AI development: tasks that humans find trivial (like intuitively mapping a sound to a shape) can reveal when an AI is doing something fundamentally different. Conversely, as AI models become more complex, using them as comparative *subjects* can also yield questions for cognitive science—pushing us to define what intuitive processing really entails, and how it might be emulated.

While our experiment delivered clear findings, there are some points to acknowledge. First, the number of nonword stimuli was relatively small (six), chosen to represent distinct extremes (very smooth *vs.* very sharp sounds). A broader set of stimuli varying along multiple acoustic dimensions would allow a more fine-grained understanding of which sound features drive the mappings and whether there are sounds that produce ambiguous or idiosyncratic associations. Second, GPT-4 was tested in a text-only manner; it would be informative to test other AI systems, including those with audio or visual capabilities. Third, in comparing a group of 40 humans to a single AI model output per stimulus, our analysis necessarily treats the AI as one virtual participant. In future work, one might simulate a distribution of AI responses by sampling multiple outputs from the model (using temperature to induce variability) or by including multiple LLMs, to more rigorously compare variability and consensus between hu-

man and AI cohorts.

Despite these limitations, the effect observed was strong and consistent enough to support our conclusions. For each of the divergent cases, it is unlikely that additional data or slight methodological tweaks would overturn the basic finding that GPT-4’s mapping differed from the majority human choice. If anything, expanding the study could further illuminate the boundary conditions: for instance, identifying if there are certain types of sounds or descriptive nuances where GPT-4 aligns more with humans, which could hint at which aspects of grounding it partly acquires from language.

5. Conclusion

This research provides a clear demonstration of the gap between human intuitive processing and the semantic, language-based reasoning of a state-of-the-art AI. In a task as straightforward to humans as associating a nonverbal sound with a visual shape, humans drew on a wealth of embodied experience and cross-modal intuition, resulting in a high degree of consensus within our participant sample. GPT-4, by contrast, approached the task armed with its vast linguistic knowledge but without true sensory grounding. The result was a systematic mismatch in how sounds were visualized, reflecting the model’s reliance on learned semantic correlations rather than direct perceptual analogues.

The findings underline that certain aspects of human cognition-especially those rooted in our embodiment and perceptual systems-remain challenging for current AI to replicate. As AI continues to progress, our study high-

lights the importance of incorporating multimodal and grounding elements into model architectures if we aim to achieve more human-like understanding. Simply training on more text may not teach a model why a “smooth” sound feels round; for that, the model may need to hear and see, and form its own cross-sensory mappings.

From an interdisciplinary standpoint, this work also demonstrates a fruitful approach to evaluating AI systems, using methods from cognitive psychology to probe the nature of AI’s “thought.” The differences we observed not only provide targets for AI improvement but also offer insight into the uniqueness of human cognition. Our intuitive melding of senses, honed by evolution and experience, endows us with a richness of understanding that purely language-trained models currently lack. Bridging that divide is an exciting frontier. Progress in this direction will not only advance AI but also deepen our understanding of the human mind, by forcing us to formally characterize those ineffable intuitive leaps that even a powerful machine finds difficult to imitate.

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